The LRL Probe: a Four-Sample Functional as Family Coordinate, Numerical Fingerprint and Growth Diagnostic

An analytical and empirical study

Sajid Hussain
TRI-RANGE INSTITUTE OF NATURAL RESOURCES AND TECHNOLOGY
Karakoram · Himalaya · Hindu Kush
Gilgit-Baltistan Pakistan
sajidglt@gmail.com

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Abstract

The LRL probe assigns to a positive function f the quantities $g(n) = \big(f(2n{+}1) - 1\big) + \big(1/f(2n{+}1) - 1/f(2n{+}3)\big) + \ln\big(f(2n{+}1)/f(2n{+}3)\big)$ and the number $\mathrm{LRL}(f) = g(0)\,g(2)/g(1)$. It has been proposed informally as a way to identify and classify mathematical functions. We give the first systematic analysis of what this number can and cannot do. On the negative side we prove that no continuous scalar can identify functions, exhibit explicit collision families, and show that LRL carries no more information than the four samples $f(1), f(3), f(5), f(7)$. On the positive side we show that within natural one-parameter families LRL is a one-to-one, numerically invertible coordinate on explicit ranges, derive exact asymptotic laws, and measure its behaviour as a hash: over 3857 distinct functions it is worth about 19 bits at six significant digits, with colliding pairs falling roughly tenfold per additional digit. We then construct an upgraded identifier, LRL-ID, that is defined for every real expression in up to six variables, is invariant to the sign and scale of an equation, and produced no collisions over 3858 single-variable functions and none among 1022 distinct multivariable equations, while recognising equivalent spellings that symbolic canonicalisation misses. As an exact companion we give a decodable integer code Z for canonical expressions. Finally we show that the ratio structure of LRL is a log-convexity test that recovers polynomial degree and exponential base of sequences, relate it to a grid-invariant "growth spectrum" classifier with a one-sided Hardy-field bound, and note that the reciprocal term of the probe reproduces the Leibniz series for π. Each component is placed against the prior work it descends from, and all results are reproducible from the accompanying code.

Contents

1 Introduction **2**

2 Related work **3**

3 Definition and elementary structure **4**

4 LRL as a coordinate inside one-parameter families **5**
4.1 Exact expressions 5
4.2 Monotone ranges and inversion 6

5 Limits of scalar identification **7**

6 A collision census **8**

7 LRL-ID: an upgraded identifier **9**
7.1 Construction 9
7.2 Why a curve and not a ray 9
7.3 Results 10

8 An exact companion: the decodable code Z **11**

9 The probe as a sequence diagnostic **11**

10 The growth-spectrum classifier **12**
10.1 Three levels 12
10.2 A one-sided bound 13
10.3 Evaluation 14
10.4 Threshold heuristics built on the probe 14

11 A classical identity: the Leibniz series and the circle **15**

12 Discussion **15**

13 Conclusion **16**

A Reference implementation **16**

B Reproducibility **17**

1 Introduction

A recurring wish in computational mathematics is a *periodic table of functions*: a small set of numbers that identifies an equation in the way atomic number identifies an element. Several rigorous partial answers exist. Coefficient sequences identify analytic functions on an interval to machine precision [39, 13]; holonomic functions are identified by a linear differential equation with polynomial coefficients [42, 32]; integer sequences are catalogued by lookup [36]; and syntactic objects can be numbered exactly in the manner of Gödel [15].

This paper studies a much simpler proposal, the *LRL probe*. For a function f that is positive on the odd integers it combines a linear, a reciprocal and a logarithmic term (Definition 3.1) and forms a single ratio. The formula reached us through informal computational frameworks that use it to "classify" functions by thresholding ratios; it has, to our knowledge, no prior analysis. Our aim is to establish precisely what the number measures.

Contributions.

1. **Structure** (Section 3). An exact decomposition of the probe in terms of the Itakura–Saito generator, closed forms and asymptotics, and a sensitivity analysis showing the number is dominated by $f(1)$ and $f(5)$.
2. **Family coordinate** (Section 4). Exact expressions and asymptotic laws for four one-parameter families, the poles and monotone ranges of LRL along each, and numerical inversion recovering the parameter to machine precision.
3. **Limits** (Section 5). A proof that no continuous scalar invariant can be injective, explicit collisions, and the observation that LRL factors through four samples.

4. **Measurement** (Section 6). A collision census over 3858 distinct functions quantifying LRL as a hash.
5. **LRL-ID** (Section 7). An upgraded three-component identifier with a positive transform, shifted lattices, a curve embedding for several variables with a provable monomial-separation property, and sign and scale invariance for equations.
6. **An exact code** (Section 8). A bijective, decodable integer Z for canonical expression trees, with its undecidability limit.
7. **Growth diagnostics** (Sections 9–10). The shape ratio of the probe as a log-convexity test, and a three-level growth-spectrum classifier with a Hardy-field one-sided bound.
8. **A classical identity** (Section 11). The reciprocal term of the probe yields the Leibniz series, giving π and the circumference of a circle.

Scope and honesty of claims. Almost every ingredient used here is classical, and Section 2 traces each one to its source. After that placement the contribution reduces to the first analysis of the LRL formula itself, one argument we could not find elsewhere (the Hardy-field bracket of Section 10), and the assembly of known parts into an identifier and a classifier with measured behaviour. Our literature search was targeted rather than systematic, so even that one claim should be read as "unclaimed in what we searched" rather than "new". Every number reported was produced by the accompanying code (Appendix B).

2 Related work

Each component of this paper descends from an identifiable line of earlier work. We list the lines here so that the contribution is stated at its true size.

Fingerprints of expressions by evaluation. Martin [26] decided the equivalence of algebraic expressions by hash-coding their values at random points. Gonnet [16] formalised such *signatures*, showing that the answer "not equivalent" is always right while "equivalent" is right with high probability; Schwartz and Zippel [33, 43] supplied the polynomial bounds. SymPy's `equals` method still works this way [37]. LRL-ID (Section 7) is a signature in exactly this sense, with a fixed rather than random evaluation scheme and the LRL formula as the compressing map. Its independence of spelling (Table 5) is the defining property of such signatures and owes nothing to the LRL formula.

Reduction of several variables to one. The curve embedding of Definition 7.2 is a real-exponent analogue of Kronecker substitution [25], which maps a multivariate polynomial to a univariate one by substituting powers of a single variable so that distinct monomials receive distinct exponents; randomised variants are studied by Arnold and Roche [2]. Using rationally independent exponents removes the possibility of monomial collision altogether, at the price of non-integer exponents.

Empirical complexity. Goldsmith, Aiken and Wilkerson fit power laws to measured cost [17]. Zaparanuks and Hauswirth infer an empirical cost function per routine by discovering the program's inputs ("algorithmic profiling") [41], and Coppa, Demetrescu and Finocchi measure how each routine scales with an automatically estimated input size ("input-sensitive profiling") [11]. The `big_O` package fits a fixed set of classes by least squares [4]; the `bigO` package selects among classes by the Akaike criterion and reports p-values [5]. The doubling test of Sedgewick and Wayne [35] is the textbook form of the local exponent. Clauset, Shalizi and Newman [10] document the unreliability of log-log slope fits for distributions; our setting, deterministic

functions on a geometric grid, avoids their tail-sampling problem, but their warning about pre-asymptotic ranges applies verbatim. Our classifier differs from the tools above in operating on local exponents rather than raw costs, in returning an interval, and in having an explicit unresolved state.

Series analysis and observed order. The exponent spectrum and its extrapolation are the *ratio method* of Domb and Sykes [12], reviewed by Guttmann [18], applied to function values instead of series coefficients; the Domb–Sykes plot of successive ratios against $1/n$ is the same object as our Level 1. The correction signature s_k is, term for term, the *observed order of accuracy* $\ln(\varepsilon_{32}/\varepsilon_{21})/\ln r$ computed from three systematically refined grids in the verification of numerical simulations [30], with the roles of grid spacing and sample index exchanged. Aitken and Richardson extrapolation [1, 29] are the standard accelerators in both fields.

Abstention. Reporting "unresolved" is a *reject option* in the sense of Chow [9]: abstain when the evidence does not reach a threshold, trading coverage for error rate. Our thresholds are deterministic tests of monotonicity and fit quality rather than posterior probabilities, but the idea is his.

Hardy fields. Hardy [19], Rosenlicht [31] and Boshernitzan [7] establish that the logarithmico-exponential functions form Hardy fields, whose elements are eventually monotone and pairwise comparable; this is the mathematical basis of every statement of the form "f grows faster than g". We did not find a prior use of the eventual monotonicity of the elasticity xf'/f as a one-sided bound in empirical exponent estimation (Proposition 10.2). Given the size of the numerical-analysis literature we regard this as unclaimed rather than established as new.

3 Definition and elementary structure

Definition 3.1 (LRL probe)**.** Let f be positive at the odd integers $1, 3, 5, \ldots$. For $n \geq 0$ put $a = f(2n+1)$, $b = f(2n+3)$ and

$$g(n) = \underbrace{(a-1)}_{L(n)} + \underbrace{\left(\frac{1}{a} - \frac{1}{b}\right)}_{R(n)} + \underbrace{\ln\frac{a}{b}}_{\Lambda(n)}, \qquad \mathrm{LRL}(f) = \frac{g(0)\,g(2)}{g(1)}. \tag{1}$$

$\mathrm{LRL}(f)$ is defined when $g(1) \neq 0$ and depends only on $f(1), f(3), f(5), f(7)$.

Example 3.2. For $f(x) = x$: $g(0) = -0.431946$, $g(1) = 1.622508$, $g(2) = 3.720671$ and $\mathrm{LRL} = -0.990521$.

Proposition 3.3 (Decomposition)**.** *With $\varphi(u) = u - 1 - \ln u$, the generator of the Itakura–Saito divergence [22, 14],*

$$g(n) = (a-1) + \varphi(1/a) - \varphi(1/b).$$

Proof. $\varphi(1/a) - \varphi(1/b) = \frac{1}{a} - 1 + \ln a - \frac{1}{b} + 1 - \ln b = \frac{1}{a} - \frac{1}{b} + \ln\frac{a}{b}$. □

Since $\varphi \geq 0$ with equality only at $u = 1$, the reciprocal and logarithmic terms together measure the difference of two "distances from one". The probe is not a divergence between a and b: the linear term $a - 1$ breaks the symmetry. The three terms respond to three regimes, and the dominant one shows the regime. For $f = 1000 + x$ the terms of $g(0)$ are 1000.0, 0.0000 and -0.0020. For $f = 1/x$ they are 0, -2.0000 and 1.0986. For $f = 1 + 0.01x$ they are 0.0100, 0.0192 and -0.0196.

Proposition 3.4 (Constants)**.** *If $f \equiv c > 0$ with $c \neq 1$ then $g(n) = c - 1$ for all n and $\mathrm{LRL}(f) = c - 1$. If $c = 1$ then $g \equiv 0$ and* LRL *is undefined.*

Proposition 3.5 (Identity function)**.** *For $f(x) = x$ and $m = 2n + 1$,*

$$g(n) = m - 1 + \frac{2}{m(m+2)} - \ln\Big(1 + \frac{2}{m}\Big) = m - 1 - \frac{2}{m} + \frac{4}{m^2} - \frac{20}{3m^3} + \frac{12}{m^4} + \mathcal{O}(m^{-5}).$$

Proof. Expand $2/(m(m+2)) = 2m^{-2} - 4m^{-3} + 8m^{-4} - \dots$ and $\ln(1+2/m) = 2m^{-1} - 2m^{-2} + \frac{8}{3}m^{-3} - 4m^{-4} + \dots$ and collect terms. □

Numerically the remainder after the cubic term, multiplied by m^4, equals 11.02, 11.89 and 10.94 at $n = 10, 100, 1000$, consistent with the coefficient 12. Hence for growing f the probe is essentially $g(n) \approx f(2n + 1)$: the reciprocal and logarithmic terms are lower-order corrections, a fact with consequences in Section 10.4.

Sensitivity. Write $\mathrm{LRL} = \Lambda(f_1, f_3, f_5, f_7)$. At the identity point $(1, 3, 5, 7)$ the gradient of Λ is

$$\nabla\Lambda = (2.2932,\ 0.2366,\ -0.4065,\ 0.0326), \qquad \Big(\frac{x_i\,\partial_i\Lambda}{\Lambda}\Big)_i = (-2.32,\ -0.72,\ +2.05,\ -0.23),$$

the second vector being the elasticities. The number is governed mainly by $f(1)$ and $f(5)$ and is almost insensitive to $f(7)$.

4 LRL as a coordinate inside one-parameter families

Although a scalar cannot identify all functions (Section 5), it may well parametrise a *family*. We treat four.

4.1 Exact expressions

Power laws $f = x^p$.

$$\begin{aligned} g(0) &= 1 - 3^{-p} - p\ln 3, \\ g(1) &= 3^p - 1 + 3^{-p} - 5^{-p} + p\ln\tfrac{3}{5}, \\ g(2) &= 5^p - 1 + 5^{-p} - 7^{-p} + p\ln\tfrac{5}{7}. \end{aligned}$$

Exponentials $f = b^x$.

$$\begin{aligned} g(0) &= b - 1 + b^{-1} - b^{-3} - 2\ln b, \\ g(1) &= b^3 - 1 + b^{-3} - b^{-5} - 2\ln b, \\ g(2) &= b^5 - 1 + b^{-5} - b^{-7} - 2\ln b. \end{aligned}$$

Lines through the origin $f = kx$.

$$\begin{aligned} g(0) &= k - 1 + \tfrac{2}{3k} - \ln 3, \\ g(1) &= 3k - 1 + \tfrac{2}{15k} - \ln\tfrac{5}{3}, \\ g(2) &= 5k - 1 + \tfrac{2}{35k} - \ln\tfrac{7}{5}. \end{aligned}$$

Theorem 4.1 (Asymptotic laws)**.** *(i)* $\mathrm{LRL}(x^p) \sim \big(1 - 3^{-p} - p\ln 3\big)(5/3)^p$ *as* $p \to \infty$, *and* $\mathrm{LRL}(x^p) = -\frac{1}{2}\ln 3\,\ln 5\,p^2 + \mathcal{O}(p^3)$ *as* $p \to 0$.

(ii) $\mathrm{LRL}(b^x) \sim b^2\,(b - 1 - 2\ln b)$ *as* $b \to \infty$, *and* $\mathrm{LRL}(b^x) = \frac{5}{3}(b-1) + \mathcal{O}((b-1)^2)$ *as* $b \to 1$.

(iii) $\mathrm{LRL}(kx) = \frac{5}{3}k + \mathcal{O}(1)$ *as* $k \to \infty$*, with poles at*

$$k_\pm = \frac{c \pm \sqrt{c^2 - 8/5}}{6}, \qquad c = 1 + \ln\tfrac{5}{3},$$

that is $k_- = 0.114106$ *and* $k_+ = 0.389503$*.*

Proof. (i) For large p, $g(1) \sim 3^p$ and $g(2) \sim 5^p$ while $g(0)$ is as displayed. For small p, $g(0) = 1 - e^{-p\ln 3} - p\ln 3 = -\frac{1}{2}(p\ln 3)^2 + \mathcal{O}(p^3)$, $g(1) = p\ln 3 + \mathcal{O}(p^2)$ and $g(2) = p\ln 5 + \mathcal{O}(p^2)$. (ii) For large b, $g(0) \sim b - 1 - 2\ln b$, $g(1) \sim b^3$, $g(2) \sim b^5$. For $b = 1 + \varepsilon$ a second-order expansion gives $g(0) = \varepsilon - 4\varepsilon^2$, $g(1) = 3\varepsilon - 5\varepsilon^2$, $g(2) = 5\varepsilon - 2\varepsilon^2$ up to $\mathcal{O}(\varepsilon^3)$. (iii) The leading terms are k, $3k$, $5k$. Poles are the zeros of $g(1)$; multiplying $g(1) = 0$ by k gives $3k^2 - ck + \frac{2}{15} = 0$. □

The laws were checked numerically: the ratio of exact to leading term is 1.0001 for x^p at $p = 10$, 1.0001 for b^x at $b = 100$, and 0.9973 for b^x at $b = 1.001$; at $p = 5 \times 10^{-4}$ the exact value -2.211×10^{-7} matches $-\frac{1}{2}\ln 3 \ln 5\, p^2 = -2.210 \times 10^{-7}$.

4.2 Monotone ranges and inversion

Figure 1 shows LRL along the four families. A horizontal line meeting a curve more than once is a collision *inside* the family. Table 1 lists the poles, the range on which LRL is strictly monotone and unbounded, and the largest number of family members sharing one value, determined on grids of 6000 parameter values.

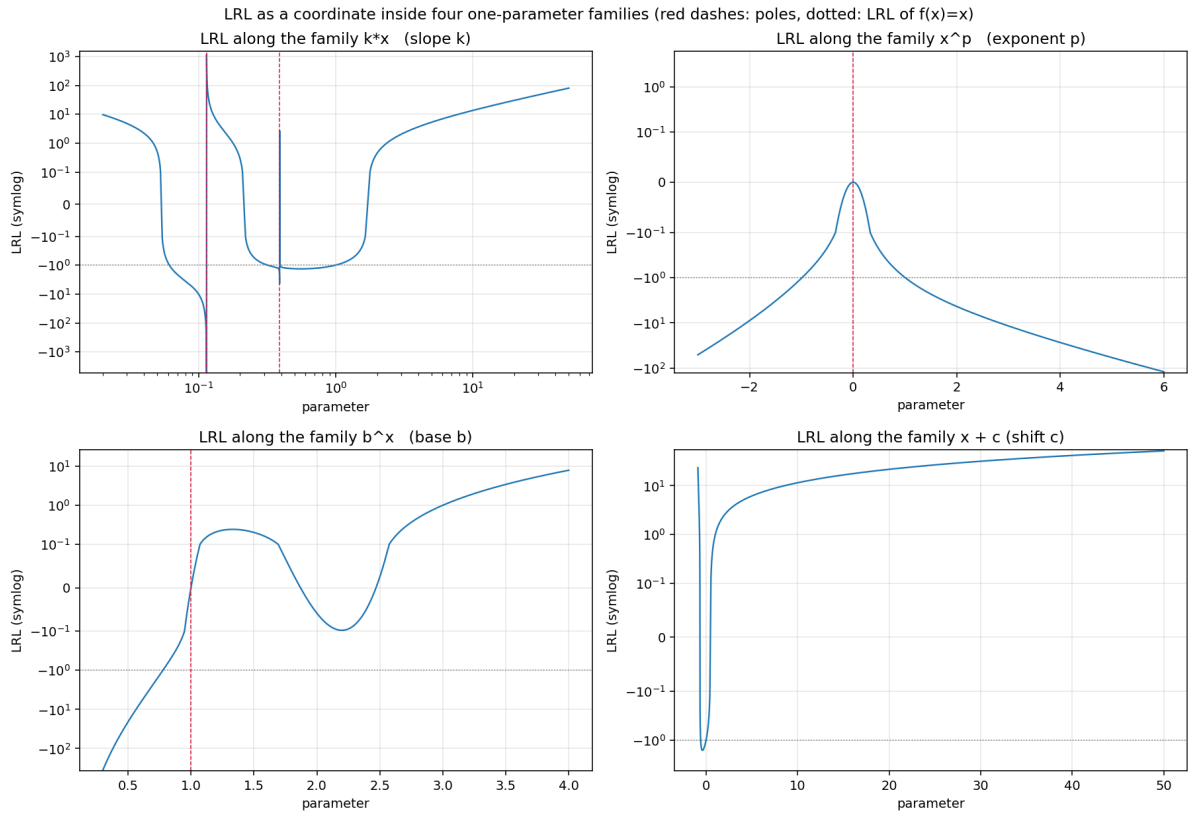


Figure 1: LRL along four one-parameter families (symmetric-log vertical axis). Red dashed lines mark poles; the dotted line is $\mathrm{LRL}(x) = -0.990521$.

Table 1: LRL as a coordinate inside four families. The last column is the largest number of family members sharing one value of LRL.

family	poles	turning points of LRL	one-to-one range	max.
kx	$k = 0.1141$ and 0.3895	minimum -1.338 at $k = 0.558$	$k \geq 0.558$	4
x^p	none (removable at $p = 0$)	maximum 0 at $p = 0$	$p > 0$; separately $p < 0$	2
b^x	none (removable at $b = 1$)	maximum 0.2410 at $b = 1.332$; minimum -0.0979 at $b = 2.198$	$b > 2.678$, where LRL > 0.2410	3
$x + c$	none	minimum -1.614 at $c = -0.399$	$c \geq -0.40$	2

On a one-to-one range the number is a genuine measurement of the parameter. Using Brent's method [8] we recovered, from the value of LRL alone: $p = 2.5$ from LRL $= -7.32948761$; $b = e$ from 0.30881121; $k = 7$ from 8.62430417; and $k = 2\pi$ from 7.43666985; each to within 10^{-15}. For power laws with $p > 0$ the LRL number therefore *orders the family by exponent.*

5 Limits of scalar identification

Theorem 5.1 (No continuous scalar identifier)**.** *Let $U \subset \mathbb{R}^m$ be open and non-empty with $m \geq 2$. There is no continuous injective map $h : U \to \mathbb{R}$.*

Proof. Let $B \subset U$ be an open ball. $h(B)$ is a connected subset of $\mathbb{R}$ containing more than one point, hence an interval I. Choose $p \in B$ with $h(p)$ in the interior of I; this is possible since I has at most two endpoints and h is injective. Then $B \setminus \{p\}$ is connected, but its image $I \setminus \{h(p)\}$ is not, contradicting continuity. □

Corollary 5.2. *Any identifier that is a continuous function of two or more function values, in particular* LRL $= \Lambda(f_1, f_3, f_5, f_7)$*, has collisions. Where $\nabla\Lambda \neq 0$ the level sets of Λ are three-dimensional hypersurfaces in $\mathbb{R}^4_{>0}$.*

The obstruction is topological and cannot be designed away. A cardinality argument closes the remaining door: the set of all functions $\mathbb{R} \to \mathbb{R}$ has cardinality $2^{\mathfrak{c}}$, strictly larger than $\mathbb{R}$, so no injective labelling by reals exists at all; for continuous functions a set-theoretic bijection with $\mathbb{R}$ exists but is necessarily discontinuous and carries no structure.

Explicit collisions.

(a) *Invisible perturbations.* If h vanishes on $\{1, 3, 5, 7\}$ then $\mathrm{LRL}(f + h) = \mathrm{LRL}(f)$. Thus x, $x + (x-1)(x-3)(x-5)(x-7)$, $x + 1000\sin(\pi x)$ all give -0.990521, although the third oscillates by thousands between integers.

(b) *Level-set collisions.* Even with different sample values the scalar repeats. In the family $k\,x^p$ the value $\mathrm{LRL}(x)$ is attained at the pairs in Table 2; in particular three different straight lines share it.

Table 2: Members of $\{k\,x^p\}$ with LRL equal to $\mathrm{LRL}(x) = -0.990521$ (to 10^{-9}).

p	values of k
0.5	0.0807853, 0.6798277
0.75	0.0749951, 0.6618156
1	0.0610725, 0.3171013, 1
1.5	0.1108550, 0.3868398, 1.7911221
2	0.0525765, 0.3300132, 2.5089635
3	0.2467021, 3.8351335

Remark 5.3 (Information bound). $\mathrm{LRL} = \Lambda \circ S$ with $S(f) = (f(1), f(3), f(5), f(7))$. Every collision of S is a collision of LRL and not conversely, so the raw sample vector is always at least as good an identifier as the number computed from it. Whatever identifying power LRL has comes from the sample points, a theme that returns in Section 7.

6 A collision census

Constructed collisions say little about typical behaviour. We therefore generated a corpus of 3907 functions from eight families (shifted power laws $ax^p + c$ with 20 exponents; quadratics and cubics with small integer coefficients; shifted exponentials $ab^x + c$; logarithmic, $x \ln$, and square-root forms; and Möbius functions $(ax + b)/(x + c)$). After removing functions that coincide at eight irrational test points, 3858 distinct functions remained, of which 3857 have a defined LRL.

Table 3: Collisions of identifiers over the census corpus ($N = 3857$; an ideal identifier needs $\log_2 N = 11.91$ bits). Columns: functions involved in a collision; colliding pairs; effective size in bits from the birthday estimate $N^2/(2 \times \text{pairs})$; and the registry size at which a first accidental collision is expected.

identifier	functions	pairs	bits	first at $N \approx$
LRL rounded to 10^{-6} (absolute)	0	0	–	–
LRL to 7 significant digits	2	1	22.8	2727
LRL to 6 significant digits	28	14	19.0	729
LRL to 5 significant digits	–	94	16.3	281
LRL to 4 significant digits	1216	808	13.2	96
LRL to 3 significant digits	2921	7861	9.9	31
six-term vector $g(0), \ldots, g(5)$	0	0	–	–
raw samples $f(1), f(3), f(5), f(7)$	0	0	–	–

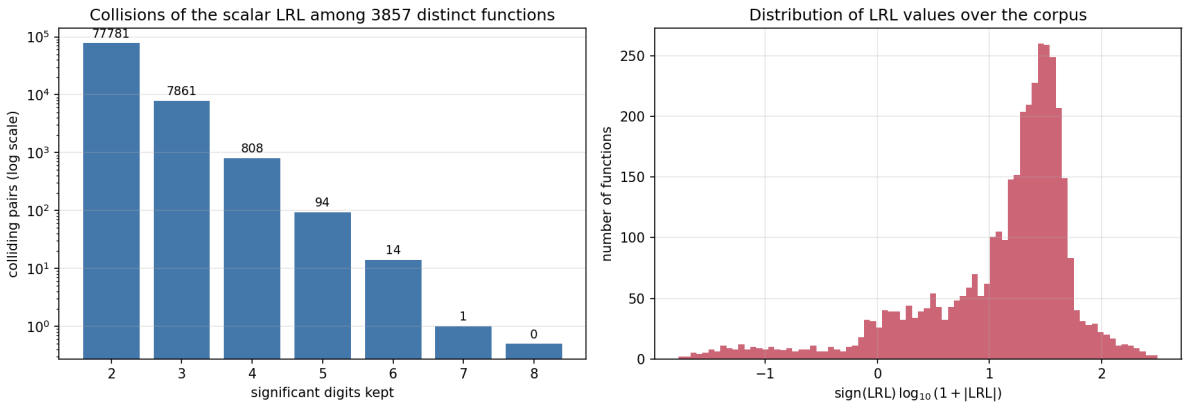


Figure 2: Left: colliding pairs of the scalar LRL against the number of significant digits retained. Right: distribution of LRL values over the corpus.

Three observations follow. First, colliding pairs fall by close to a factor of ten per retained digit (77781, 7861, 808, 94, 14, 1, 0 for 2 to 8 digits), which is the behaviour of values drawn from a continuous distribution: each digit adds about $\log_2 10 \approx 3.3$ bits. In this sense LRL behaves like a *random hash* on natural expressions. Second, at six significant digits its effective size is about 5.3×10^5 values, so by the birthday bound a registry can expect its first accidental collision near 730 entries. Third, the collisions that occur are between unrelated forms, for example $x^{2.25} + 5$ with $\sqrt{x+2} + 10$, and $x^2 + 7x + 3$ with $x^3 + 2x + 4$; nothing structural links them. The median relative gap between neighbouring LRL values is 8.4×10^{-4}.

7 LRL-ID: an upgraded identifier

The scalar has three structural gaps: it is undefined when $f \le 0$ at a sample point, which excludes most equations lhs $-$ rhs $= 0$; it samples only integers; and it is restricted to one variable. We close each gap while keeping the LRL formula.

7.1 Construction

Lemma 7.1 (Positive transform)**.** $P(y) = y + \sqrt{y^2+1} = \exp(\operatorname{asinh} y)$ *is an increasing bijection* $\mathbb{R} \to (0, \infty)$ *with* $P(-y) = 1/P(y)$.

Definition 7.2 (LRL-ID)**.** Fix shifts $\delta \in \{0,\ \phi^{-1},\ \sqrt{2}-1\}$ with ϕ the golden ratio, giving the lattices $X_\delta = \{1+\delta, 3+\delta, 5+\delta, 7+\delta\}$ and twelve distinct sample points. Fix exponents $\alpha = (1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{11})$ and constants $c = (1,\ e/2,\ \pi/3,\ \ln 5,\ 2^{1/3},\ \phi)$. For an expression F in variables $x_1 < \cdots < x_v$ (alphabetical order, $v \le 6$) let

$$f(t) = F\big(c_1 t^{\alpha_1}, \ldots, c_v t^{\alpha_v}\big).$$

If $F = 0$ is an *equation*, replace the twelve samples $f(t_i)$ by $f(t_i)/s$ where $|s|$ is their root mean square and s has the sign of the first non-zero sample. The identifier is

$$\text{LRL-ID}(F) = \Big(\Lambda\big(P(f(t)) : t \in X_\delta\big)\Big)_\delta,$$

three numbers reported to six significant digits.

7.2 Why a curve and not a ray

Two instructive failures preceded this design. On the ray $(t, \sqrt{2}\,t, \sqrt{3}\,t)$ the Pythagorean form $a^2 + b^2 - c^2$ vanishes identically, because $1 + 2 = 3$. More fundamentally, on *any* ray a homogeneous polynomial of degree d restricts to $C\,t^d$, so after scale normalisation every homogeneous equation of one degree receives the same identifier; we observed $a^2 + b^2 = c^2$ and $v^2 = 2gh$ colliding in exactly this way. The curve of Definition 7.2 removes both defects.

Proposition 7.3 (Monomial separation)**.** *Let $F \ne G$ be polynomials in at most six variables whose combined set of monomials has at most twelve elements. Then F and G differ at one or more of the twelve sample points of the curve. If $F = 0$ and $G = 0$ are equations with identical normalised samples, then $G = \lambda F$ for some $\lambda \ne 0$.*

Proof. A monomial $x^a = \prod x_j^{a_j}$ restricts to $\big(\prod c_j^{a_j}\big)\, t^{\langle a, \alpha\rangle}$. The numbers $1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{11}$ are linearly independent over $\mathbb{Q}$ [3], so distinct exponent vectors $a \in \mathbb{N}^v$ give distinct real exponents $\langle a, \alpha\rangle$. Hence $H = F - \lambda G$ restricts to a generalised polynomial $\sum_{i=1}^{m} h_i\, t^{e_i}$ with distinct exponents and $m \le 12$ terms. By Laguerre's extension of Descartes' rule of signs [23] such a function, if not identically zero, has at most $m - 1 \le 11$ positive zeros. Vanishing at twelve points forces all $h_i = 0$, i.e. $F = \lambda G$ (with $\lambda = 1$ for expressions). □

Remark 7.4. Proposition 7.3 is a statement about the twelve *samples*. The identifier compresses them to three rounded numbers, so by Remark 5.3 the guarantee does not transfer; the separation of the identifier itself is an empirical matter, reported next. The principle is that of probabilistic identity testing [33, 43]: evaluation points must be generic. In the terminology of Martin and Gonnet [26, 16], LRL-ID is a signature with deterministic evaluation points; the census below measures how much of a signature's collision resistance survives the compression to three rounded numbers.

7.3 Results

Table 4: Scalar code versus LRL-ID on cases the scalar cannot handle.

input	scalar code $\lfloor 10^6 \mathrm{LRL} \rceil$	LRL-ID
x	-990521	$(1.35525,\ 3.23613,\ 2.6353)$
$x + 1000\sin(\pi x)$	-990521	$(1.35525,\ 2.99838,\ 2.99844)$
$0.3171013\,x$	-990521	$(0.280031,\ 0.705907,\ 0.565085)$
$\ln x$	undefined	$(-0.580226,\ 0.405752,\ 0.10472)$
$x^2 - 5x + 6 = 0$	undefined	$(-0.294132,\ -0.188286,\ -0.266953)$
$y = x^2$	undefined	$(0.0370485,\ -0.101598,\ -0.0456306)$
$a^2 + b^2 = c^2$	undefined	$(0.16765,\ 0.150174,\ 0.160695)$
$v^2 = 2gh$	undefined	$(0.0484861,\ 0.0273081,\ 0.0419941)$
$V = IR$	undefined	$(-0.126193,\ -0.065709,\ -0.0866184)$
$e^{i\pi} + 1 = 0$	undefined	(∞, ∞, ∞)

Collision resistance. Over the census corpus LRL-ID is defined for all 3858 functions with no collisions. For several variables we generated 1076 equations: homogeneous quadratic forms $px^2 + qy^2 + rz^2 + sxy = 0$, sums $z = px^m + qy^n$ and monomial laws $z = kx^m y^n$. Exactly 1022 of these are distinct as equations (the rest are proportional copies), and they received exactly 1022 identifiers: no false collisions and no missed duplicates.

Invariance. $y = x^2$, $x^2 = y$ and $0 = 5x^2 - 5y$ share one identifier. An identity collapses to the marker (∞, ∞, ∞), since its samples vanish. A perturbation is still resolved: $v^2 = 2gh + 0.001$ gives $(0.0485213,\ 0.0273292,\ 0.0420186)$.

Robustness to spelling. Because the identifier is numerical it agrees on forms that symbolic simplification fails to unify (Table 5); the exact code Z of Section 8 does not.

Table 5: Equivalent spellings: exact code Z versus LRL-ID. The equalities hold for positive arguments, which is where the identifier samples.

form A	form B	same Z	same LRL-ID
$\ln(x^2)$	$2\ln x$	no	yes
$\sqrt{x^2}$	x	no	yes
$\ln x + \ln(x+1)$	$\ln(x^2 + x)$	no	yes
$(x^3)^{1/3}$	x	no	yes
$\sin 2x$	$2\sin x \cos x$	yes	yes
$e^{\ln x + 1}$	$e\,x$	yes	yes

8 An exact companion: the decodable code Z

A numerical fingerprint suggests equality but cannot prove it. For exact identification we pair it with a syntactic code in the tradition of Gödel numbering [15].

Construction. An input is parsed and brought to a canonical form by expansion and simplification [27]; for an equation lhs = rhs we take the primitive part of lhs − rhs with a fixed sign. The canonical tree is serialised in prefix order into tokens from an alphabet of size $K = 104$ (decimal digits, a terminator, sign and rational markers, letters, eight named constants, 28 function heads and an escape for unknown heads), each compound node being followed by its arity. If the token string is $(\tau_1, \dots, \tau_L)$ with $\tau_i \in \{0, \dots, K-1\}$ we set

$$Z = \sum_{i=1}^{L} (\tau_i + 1)\, K^{\,L-i}.$$

Proposition 8.1. *The map from canonical expressions to Z is injective and effectively invertible.*

Proof. Bijective base-K numeration is a bijection between finite strings over a K-letter alphabet and the non-negative integers. Prefix serialisation with explicit arities and terminated numerals is uniquely decodable, so the composition is injective and decoding reconstructs the tree. □

Z grows linearly with the expression, $\log_{10} Z \approx 2.02\, L$. Examples: $\ln x$ has $Z = 973568822523$; $n!$ has $Z = 1034401466603$; $2x$ has 17 digits, $x^2 + 2x + 1$ has 43, $E = mc^2$ has 57 and $a^2 + b^2 = c^2$ has 77. Decoding returned the canonical form in every case tested. In the periodic-table analogy we attach to Z a *mass* (weighted node count), a *period* (tree depth), a *group* (the growth class of Section 10) and a sequential *registry number*; different written forms of one canonical expression, such as $x + x$, $2x$ and $x(1 + 1)$, play the role of isotopes.

The limit. By Richardson's theorem [28] equality of expressions built from rationals, π, $\ln 2$, x, sin, exp and $|\cdot|$ is undecidable. No canonicalisation can therefore send every pair of equal expressions to one Z; Table 5 shows instances. The two identifiers are complementary: Z is exact and decodable but only as good as simplification, LRL-ID is approximate but independent of spelling. Entries with different Z and equal LRL-ID are *probable duplicates* to be confirmed. On a test sheet of 16 formulas, Z found three duplicate pairs and LRL-ID a further three.

9 The probe as a sequence diagnostic

Apply the probe to a sequence a_n through $f(n) = a_n$, and extend the ratio along the index.

Definition 9.1 (Shape ratio)**.** $C_k = \dfrac{g(k)\, g(k+2)}{g(k+1)^2}$, so that LRL $= C_0\, g(1)$.

$C_k < 1, = 1, > 1$ means the probe sequence is locally log-concave, geometric, or log-convex [38].

Proposition 9.2. *If $g(k) \sim A(2k+1)^p$ then $C_k = \left[1 - 4/(2k+3)^2\right]^p = 1 - \dfrac{4p}{(2k+3)^2} + \mathcal{O}(k^{-4})$, so $\hat{p}_k = \frac{1}{4}(1 - C_k)(2k+3)^2 \to p$. If $g(k) \sim A\, b^{2k}$ then $C_k \to 1$ and $g(k+1)/g(k) \to b^2$.*

The degree estimator is a form of Raabe's test [24]. Table 6 reports the diagnostics at $k = 20$.

Table 6: Probe diagnostics of standard sequences at $k = 20$.

sequence	LRL	C_{20}	verdict	$\hat{p}_{20}$	$g(21)/g(20)$
n	-0.9905	0.997719	polynomial-like	1.05	1.0501
triangular	-3.0713	0.995759	polynomial-like	1.96	1.0989
n^2	-4.3330	0.995670	polynomial-like	2.00	1.1000
n^3	-11.713	0.993524	polynomial-like	2.99	1.1536
$n\ln(n+1)$	-2.6732	0.997103	polynomial-like	1.34	1.0623
harmonic H_n	-0.2540	0.999162	polynomial-like	0.39	1.0144
Fibonacci	-1.5945	1.000000	geometric	–	$2.6180 = \phi^2$
2^n	-0.0586	1.000000	geometric	–	$4.0000 = 2^2$
Catalan	-15.295	1.003143	faster than geometric	–	$14.92\ (\to 16)$
$n!$	-51.086	1.096346	faster than geometric	–	$1806 = 43 \cdot 42$
primes p_n	1.1089	irregular	no smooth law	–	1.0676

The degree is recovered for polynomial sequences, the base for geometric ones, and irregularity is visible: for the primes C_k runs $0.957, 1.044, 1.029, 0.907, 0.996, 1.104, 0.924$ for $k = 8, \ldots, 14$, against a smooth climb $0.978, \ldots, 0.992$ for n^2. Two cautions apply. Bounded or decaying sequences saturate: for $a_n = 1/n$ the probe tends to a constant and $C_k \to 1$ with step ratio 1.001, which is not geometric growth. Small values bias the degree: $\sqrt{n}$ gives 0.66 rather than 0.5 because the "-1" of the linear term is not negligible.

10 The growth-spectrum classifier

The diagnostics of Section 9 are biased by shifts, scales and lower-order terms. We now describe a classifier that is free of these defects. Its ingredients are classical and Section 2 names their sources; what is added is the combination, the interval output, and the one-sided bound of Section 10.2.

10.1 Three levels

Sample $y_k = f(n_k)$ on a geometric grid $n_k = n_0 r^k$.

$$\text{Level 1 (exponent spectrum)} \quad \alpha_k = \frac{\ln(y_{k+1}/y_k)}{\ln r},$$

$$\text{Level 2 (correction signature)} \quad d_k = \alpha_{k+1} - \alpha_k, \qquad s_k = \frac{d_{k+1}}{d_k}.$$

α_k is the local log-log slope: it equals p exactly for $f = Cn^p$ on *any* geometric grid, which is the property that threshold-on-ratio heuristics lack (Section 10.4).

Proposition 10.1 (Signature limits)**.** *(i)* $f = Cn^a$*:* $d_k = 0$.

(ii) $f = Cn^a(1 + \delta n^{-s})$, $s > 0$*:* $s_k \to r^{-s}$.

(iii) $f = Cn^a(\ln n)^b$, $b \neq 0$*:* $\alpha_k = a + b\,\dfrac{\ln(1 + 1/(k+\kappa))}{\ln r}$ *with* $\kappa = \ln n_0 / \ln r$*, and* $s_k \to 1$ *from below, with* $s_k = \dfrac{k+\kappa}{k+\kappa+2}\big(1 + \mathcal{O}(k^{-1})\big)$.

(iv) $f = Ce^{cn}$*:* $\alpha_k = c\,n_k(r-1)/\ln r$ *and* $s_k = r$ *exactly.*

(v) $f = n!$*:* $s_k > r$ *and* $s_k \downarrow r$.

Proof. (ii) $\alpha_k = a + \big[\ln(1+\delta r^{-s} n_k^{-s}) - \ln(1+\delta n_k^{-s})\big]/\ln r = a - \delta(1-r^{-s}) n_k^{-s}/\ln r + \mathcal{O}(n_k^{-2s})$, a geometric sequence of ratio r^{-s}. (iii) $\ln n_k = (k+\kappa)\ln r$, so $\ln\ln n_{k+1} - \ln\ln n_k = \ln(1+1/(k+\kappa))$. (iv) $\ln y_{k+1} - \ln y_k = c\, n_k (r-1)$, so $\alpha_k \propto n_k$ and $d_k \propto n_k$. (v) follows from Stirling's formula, $\alpha_k = n_k(\ln n_k + \text{const})/\ln r + \dots$. □

Figure 3 shows these behaviours. Observed signatures were 0.50 for $3n^2 + 50n + 7$; 0.71 for $n^2 + n^{1.5}$ (theory $2^{-1/2}$); 0.41 for $n^{1.8} + 3\sqrt{n}$ (theory $2^{-1.3}$); 0.84, 0.85, 0.86 for $n \ln n$; exactly 2.00 for exponentials; and 2.44, 2.36, 2.30 for $n!$.

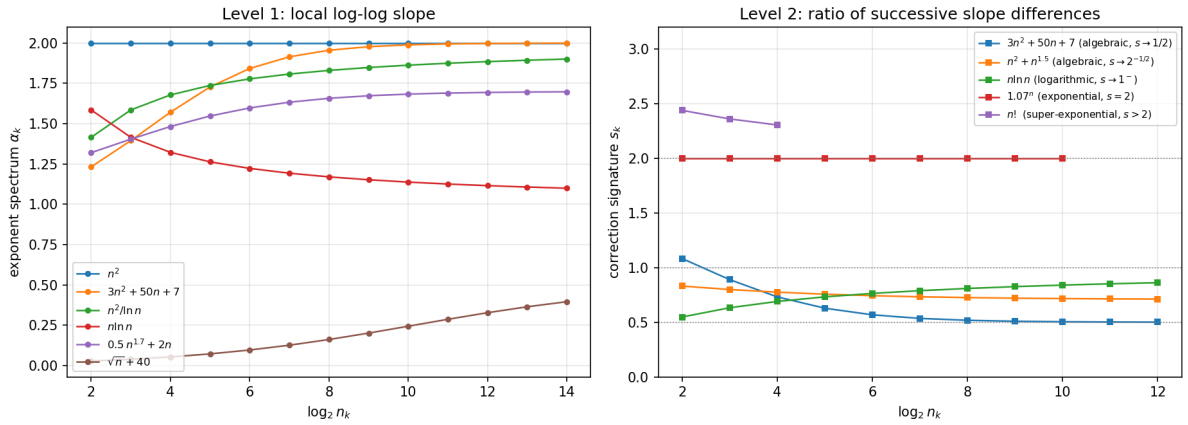


Figure 3: Left: exponent spectra converge to the true exponent from one side. Right: the correction signature separates algebraic, logarithmic, exponential and super-exponential behaviour without fitting a model.

Accelerating the signature. When a correction is still large its signature has not reached its limit. Applying Aitken's Δ^2 process [1] to the last three values repairs this: for $\sqrt{n} + 40$ the raw values 0.95, 0.90, 0.86 accelerate to 0.70, against the true $2^{-1/2} = 0.7071$. The exponent is then extrapolated by a tail model matched to the signature (constant, geometric, logarithmic, exponential, or Stirling-type), selected by the Bayesian information criterion [34]. Such extrapolation of local exponents is the ratio method of series analysis [12, 18, 29], and the signature itself is the observed order of accuracy of grid-convergence studies [30].

10.2 A one-sided bound

Proposition 10.2 (Hardy-field bracket)**.** *Let $f > 0$ be differentiable with elasticity $\varepsilon(x) = xf'(x)/f(x)$. Then*

$$\alpha_k = \frac{1}{\ln r} \int_{\ln n_k}^{\ln n_{k+1}} \varepsilon(e^u)\, du.$$

If ε is non-decreasing on $[n_K, \infty)$ with limit $a_\infty \le \infty$, then $(\alpha_k)_{k \ge K}$ is non-decreasing and $\alpha_k \le a_\infty$; symmetrically if ε is non-increasing. If f belongs to a Hardy field containing x, then ε is eventually monotone.

Proof. $\ln f(rn) - \ln f(n) = \int_n^{rn} \varepsilon(x)\, dx/x$ and the substitution $x = e^u$ give the formula; averages of a monotone function over consecutive windows of equal length are monotone and bounded by its limit. A Hardy field is a field of germs closed under differentiation, so $\varepsilon = xf'/f$ lies in it, and every element of a Hardy field is eventually monotone because its derivative has eventually constant sign [19, 7]. □

Functions built from polynomials, exp and ln generate a Hardy field. For them an observed increasing tail gives $a_\infty \ge \alpha_{\text{last}}$ and the extrapolation supplies the other end, so the classifier

reports an *interval* for the exponent and labels the class only when a single lattice point $(a, b) \in \{\dots, (1,0), (1,1), (2,0), \dots\}$ of the family $n^a(\ln n)^b$ lies in it. The assumption is that the sampled range has reached the monotone regime; it is checked on the data, and its failure triggers a refusal:

- a non-monotone tail is reported as *oscillatory or noisy*, with the candidate classes whose predicted exponent falls in the observed range;
- a signature above 1 in a non-exponential setting, or a poor tail fit, is reported as *exponent still drifting*, with the one-sided bound only.

These refusals are Chow's reject option [9] applied to complexity classification.

10.3 Evaluation

Table 7: Classifier results. "Flagged" means unresolved with the true class among the listed candidates.

test set	correct	flagged	wrong
14 textbook functions, including $n!$, $n^2/\ln n$, $n\ln^2 n$, $3n^2+50n+7$	14	0	0
15 adversarial cases: shifts, scales, bases 1.1 to 4, mixed terms, n^2-100n	14	1	0
4 oscillatory functions	0	4	0
total (33)	28	5	0

Labels were identical on four grids $(n_0, r) \in \{(4,2), (3,3), (10,2), (5,4)\}$ for eleven functions. The one adversarial case flagged is $\sqrt{n} + 400$, where the constant dominates up to $n = 32768$; the classifier reports the exponent as at least 0.14 and still rising, which is the correct description of the data. The design goal is visible in the last column: the classifier never asserts a wrong class.

10.4 Threshold heuristics built on the probe

We examined three implementations that accompany the LRL formula in circulation and classify functions by comparing ratios with fixed thresholds.

Table 8: Measured behaviour of three threshold-based implementations.

implementation	finding
consecutive-ratio classifier with a "boost"	7 of 10 textbook functions mislabelled on the grid $5, 10, \dots, 25$ (e.g. $n \mapsto \sqrt{n}$ class, $n^2 \mapsto n$ class). Reported confidence is a constant chosen by the function's name plus a fixed increment of 0.30.
probe-ratio classifier with tuned bands	56% on its own 16-function suite; $50, 56, 38, 31, 38\%$ as the sampling range changes. The bands coincide with $1+p/n$ at $n \approx 26$. Probe ratio and plain ratio $f(2n{+}3)/f(2n{+}1)$ differ by 8×10^{-4} to 9×10^{-16}, so the three terms add nothing to the ratio.
"consciousness" scoring with a physics extension	18 of 19 inputs, and random noise at the maximum score, are labelled conscious; one analysis layer raises an exception for every input and returns a default; an emission probability equals 1 at all frequencies because of a unit-scale guard; headline results are literal strings.

The common flaw of the first two is that a consecutive ratio $f(n+\Delta)/f(n) \approx 1 + p\Delta/n$ depends on the grid, whereas α_k does not. By Proposition 3.5 the probe is $g(n) \approx f(2n+1)$, so wrapping f in the probe before taking ratios changes nothing. The third makes claims its computation does not produce. Related empirical approaches that fit complexity models by least squares are [17, 4, 35].

11 A classical identity: the Leibniz series and the circle

Proposition 11.1. *For $f(x) = x$ the reciprocal term is $R(n) = \dfrac{1}{2n+1} - \dfrac{1}{2n+3}$ and*

$$\sum_{k=0}^{\infty} R(2k) = \sum_{k=0}^{\infty} \Big(\frac{1}{4k+1} - \frac{1}{4k+3} \Big) = \frac{\pi}{4}.$$

Hence the circumference of a circle of radius ρ is $2\pi\rho = 8\rho \sum_{k\geq 0} R(2k)$.

Proof. The sum is the Leibniz series $1 - \frac{1}{3} + \frac{1}{5} - \dots$ grouped in consecutive pairs; its partial sums form a subsequence of the partial sums of a convergent series [24]. □

The tail is $\sum_{k\geq N} 2/((4k+1)(4k+3)) \approx 1/(8N)$, so N pairs give π with error $\approx 1/(2N)$: measured errors are 5.0×10^{-2}, 5.0×10^{-4} and 5.0×10^{-6} at $N = 10, 10^3, 10^5$. The error has an asymptotic expansion in odd powers of $1/N$ [6], so Richardson extrapolation [29] on doubled N with factors $2, 8, 32, \dots$ is effective: four levels ($N \leq 32$) give error 1.3×10^{-9} and six levels ($N \leq 128$) give 4.4×10^{-16}.

A second route uses Section 4: the circumference $C(\rho) = k\rho$ is a line through the origin, and $k = 2\pi$ lies in the one-to-one range, so the LRL of four measured circumferences at radii $1, 3, 5, 7$, namely 7.4366698539, inverts to $k = 6.283185307179586$. This needs no formula for π but is not statistically efficient: with 1% noise on each measurement the inverted slope has relative spread 1.74%, against 0.66% for least squares through the origin (2000 trials).

We stress that $\mathrm{LRL}(x) = -0.990521$ itself contains no π; the identity concerns the reciprocal term summed over all even n.

12 Discussion

What the LRL number is good for. (1) A fast fingerprint: four evaluations, no symbolic algebra, applicable to tabulated data and black-box code; about 5 microseconds per evaluation in our implementation against roughly half a second for a symbolic canonicalisation. (2) A certain rejection test: different values prove different functions. (3) A family coordinate that can be inverted on explicit ranges and that, unlike a growth exponent, distinguishes x, $2x$ and $10x$ (-0.9905, 0.4478, 13.6058). (4) A log-convexity diagnostic for sequences.

What it is not. It is not an identifier of functions: no continuous scalar is (Theorem 5.1). It holds no information beyond four samples (Remark 5.3); the specific combination of linear, reciprocal and logarithmic terms has no privileged role in identification, and any generic function of generic samples would hash comparably. Its one-to-one ranges are limited (Table 1), it has poles, and it amplifies noise.

Recommended practice. Use the exact code Z as primary key and LRL-ID as a numerical fingerprint beside it, in the way a table of elements carries both atomic number and a measured property. Treat equal LRL-ID with different Z as a probable duplicate requiring confirmation; the matches in Table 5 hold on the positive sample domain, which is exactly why confirmation is needed.

Limitations of this study. The census corpus is synthetic and drawn from eight families; collision rates on other populations may differ. Identifiers are computed in double precision. Canonical forms depend on one computer algebra system [27]. The classifier evaluation uses exact function values; measured runtimes carry noise and frequently, and correctly, produce the unresolved state. A targeted literature search (Section 2) found direct precedents for the fingerprint, the curve embedding, the signature and the refusal state; it found none for the Hardy-field bracket, but it was not exhaustive.

Open questions. Is there a closed form for the monotone threshold of $\mathrm{LRL}(b^x)$? Can Proposition 7.3 be extended from samples to a provable collision bound for a suitably *randomised* identifier, in the spirit of [33]? Can the Hardy bracket be made fully rigorous from finitely many samples under quantitative assumptions on ε'?

13 Conclusion

The LRL number is neither the universal label it has been hoped to be nor an empty quantity. It is a lossy function of four samples that behaves like a hash of about nineteen bits on natural expressions, is an exact and invertible coordinate inside families, and encodes a log-convexity test. Its weaknesses as an identifier are structural, and are repaired not by the formula but by the choice of sample points: a positive transform, non-integer lattices and a curve with rationally independent exponents yield an identifier that separated every distinct function and equation we tested and sees through differences of spelling. Paired with a decodable exact code, it gives a practical two-number identity for equations, which is as close to a periodic table of functions as the mathematics allows.

A Reference implementation

```
import math

def g(f, n):
    a, b = f(2*n + 1), f(2*n + 3)
    return (a - 1) + (1/a - 1/b) + math.log(a / b)

def lrl(f):
    return g(f, 0) * g(f, 2) / g(f, 1)          # lrl(lambda x: x) = -0.990521...

PHI    = (1 + 5**0.5) / 2
SHIFTS = (0.0, 1/PHI, 2**0.5 - 1)
CONST  = (1.0, math.e/2, math.pi/3, math.log(5), 2**(1/3), PHI)
ALPHA  = (1.0, 2**0.5, 3**0.5, 5**0.5, 7**0.5, 11**0.5)

def lrl_id(F, nvars, is_equation=False, digits=6):
    f = lambda t: F(*[CONST[j] * t**ALPHA[j] for j in range(nvars)])
    raw = [[f(2*n + 1 + d) for n in range(4)] for d in SHIFTS]
    if is_equation:                              # sign and scale invariance
        flat = [v for row in raw for v in row]
        s = math.sqrt(sum(v*v for v in flat) / len(flat))
        if s == 0:
            return (math.inf,) * 3               # an identity
        s = math.copysign(s, next(v for v in flat if v != 0))
        raw = [[v / s for v in row] for row in raw]
    P, out = (lambda y: y + math.sqrt(y*y + 1)), []
    for row in raw:
```

```
        v = [P(y) for y in row]
        gg = [(v[i] - 1) + (1/v[i] - 1/v[i+1]) + math.log(v[i] / v[i+1])
              for i in range(3)]
        if gg[1] == 0:
            out.append(math.inf)
        else:
            out.append(float(f"{gg[0] * gg[2] / gg[1]:.{digits}g}"))
    return tuple(out)
```

B Reproducibility

All experiments were run with Python 3.12.7, NumPy 1.26.4 [20], SciPy 1.13.1 [40], SymPy 1.14.0 [27] and Matplotlib 3.9.2 [21] on 64-bit Windows. The accompanying archive contains:

- `growth_spectrum.py`: the classifier of Section 10 and its regression suite (Table 7);
- `equation_registry.py`: canonicalisation, the code Z and its decoder (Section 8);
- `lrl_numbers.py`: the scalar code, the six-term vector and LRL-ID (Section 7);
- `lrl_circle.py`: Section 11;
- `lrl_dive_families.py`, `lrl_dive_census.py`, `make_figures.py`: Sections 4 and 6 and all figures;
- the three implementations of Table 8 with the scripts that tested them;
- the captured text output of every script.

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